B3.2 Transport in plants

NAME	DATE	TOTAL
		_____ / 33

About 35 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

6 MARKS

1 STATE the polymer that strengthens xylem walls. [B3.2.8] [1]

2 DISTINGUISH between cohesion and adhesion. [B3.2.7] [2]

3 STATE the function of pits in xylem vessels. [B3.2.8] [1]

4 STATE where the xylem is found in a cross-section of a dicot root, and of a dicot stem. [B3.2.9, 10] [2]

Section B: Understanding

13 MARKS

5 EXPLAIN how transpiration causes water to move up the stem. [B3.2.7] [4]

6 EXPLAIN how the structure of xylem vessels suits them to transport water. [B3.2.8] [3]

7 DRAW a labelled plan diagram of a cross-section of a dicot stem, and annotate the function of two tissues. [B3.2.9] [4]

8 A student wrote: "The roots pump water up the stem using ATP." EXPLAIN why this is not the main mechanism. [B3.2.7] [2]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

9 [B3.2.7]

The diameter of a tree trunk and the rate of transpiration were measured over one day.

TIME OF DAY	06:00	10:00	14:00	18:00	22:00
TRUNK DIAMETER / MM	204.0	203.4	203.0	203.4	203.9
TRANSPIRATION / ARBITRARY UNITS	5	40	70	30	4

a DESCRIBE the relationship between the transpiration rate and trunk diameter. [1]

b CALCULATE the percentage decrease in diameter from 06:00 to 14:00. [2]

c EXPLAIN how these data support the cohesion-tension theory. [3]

10 [B3.2.7, B3.2.9]

A leafy celery stalk stood in dye for 2 hours. It was cut at intervals, and the highest point reached by the dye was recorded.

CONDITION	DISTANCE MOVED BY DYE / MM
LEAVES PRESENT	96
LEAVES REMOVED	24
LEAVES COATED IN PETROLEUM JELLY	30

a CALCULATE the mean rate of dye movement with leaves present, in mm per minute. [1]

b EXPLAIN the effect of removing the leaves. [2]

c EXPLAIN the effect of coating the leaves in petroleum jelly. [2]

d PREDICT where the dye would be seen in a cross-section of the stalk. [1]

e SUGGEST two improvements to this method. [2]